

EL TOR, Egypt

WMO: 624590

Lat: 28.209N

Lon: 33.646E

Elev: 35

StdP: 100.90

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.0	10.1	-3.1	2.9	14.4	-1.3	3.4	14.0	14.3	18.1	13.5	18.5	4.1	10	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	36.9	22.2	35.2	22.4	34.0	22.6	26.2	30.9	25.7	30.6	25.3	30.3	6.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.0	29.3	24.2	19.2	29.2	23.8	18.7	29.1	81.6	31.0	79.7	30.6	77.8	30.3	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	13.4	12.6	DB	6.4	41.1	1.4	1.5	5.4	42.2	4.6	43.1	3.8	44.0	2.8	45.1	
			WB	2.9	27.6	1.2	1.5	2.1	28.8	1.4	29.6	0.7	30.5	-0.1	31.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.6	15.7	16.9	19.8	22.8	26.3	28.4	29.8	30.4	28.6	25.3	21.0	17.3
	DBStd	5.48	1.92	2.26	2.67	2.57	2.24	1.91	1.53	1.39	1.59	1.91	2.07	1.91
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	196	84	51	13	1	0	0	0	0	0	0	3	43
	CDD10.0	4955	177	194	303	385	506	553	614	632	558	474	331	227
	CDD18.3	2108	3	12	58	136	248	303	356	374	308	216	83	12
	CDH23.3	22652	17	56	292	829	2221	3401	4511	4999	3823	2013	449	43
CDH26.7	9326	1	6	69	240	793	1421	2135	2544	1586	487	44	1	
Wind	WSAvg	6.5	5.3	5.5	6.3	6.5	7.0	8.1	7.5	8.1	7.7	5.7	4.9	5.0
Precipitation	PrecAvg	8	2	0	1	0	0	0	0	0	0	1	2	1
	PrecMax	30	17	7	17	11	2	0	0	0	1	12	30	13
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	7	3	1	4	2	0	0	0	0	0	3	6	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	27.6	31.4	34.3	37.7	38.9	39.1	39.1	36.3	33.2	29.7	26.1
		MCWB	17.6	17.9	18.1	19.1	20.6	21.5	22.4	22.5	22.4	22.0	21.0	18.7
	2%	DB	23.1	25.0	28.6	31.5	34.3	36.0	36.1	36.1	34.1	31.0	27.9	24.5
		MCWB	16.6	16.9	17.7	18.6	20.3	21.4	22.6	23.4	22.7	22.7	20.7	17.4
	5%	DB	21.7	23.2	26.3	29.3	32.4	33.9	34.4	34.6	32.6	29.8	26.8	23.3
		MCWB	15.4	16.0	17.2	18.3	20.2	21.8	23.0	23.5	23.2	22.7	20.1	16.5
	10%	DB	20.6	21.9	24.6	27.5	30.5	32.2	33.1	33.3	31.6	29.0	25.8	22.2
		MCWB	14.3	15.3	16.8	17.9	20.1	21.9	23.2	23.8	23.6	22.5	19.4	15.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	20.3	21.2	21.8	24.1	25.1	26.3	27.2	26.6	25.7	23.1	20.6
		MCDB	23.3	25.0	26.6	27.7	30.8	30.7	31.3	31.6	30.9	29.2	27.3	24.8
	2%	WB	17.7	18.4	19.6	20.8	23.0	24.3	25.7	26.4	25.8	24.9	22.1	18.7
		MCDB	21.9	23.1	25.3	27.2	29.0	30.1	30.9	31.1	30.2	28.8	26.8	23.4
	5%	WB	16.1	17.2	18.6	20.1	22.2	23.8	25.2	25.9	25.3	24.2	21.1	17.3
		MCDB	21.1	22.1	24.2	26.0	28.3	29.7	30.5	30.8	30.0	28.2	25.9	22.6
	10%	WB	14.8	15.9	17.7	19.4	21.6	23.3	24.8	25.5	24.9	23.6	20.0	16.1
		MCDB	20.1	21.2	23.3	25.3	27.7	29.3	30.1	30.6	29.8	27.8	25.1	21.7
Mean Daily Temperature Range	5% DB	MDBR	9.4	9.0	8.4	8.4	8.5	8.4	7.9	7.2	6.2	7.1	8.8	9.5
		MCDBR	10.7	10.9	11.5	12.0	12.2	11.5	10.1	9.2	8.2	7.9	9.6	10.4
		MCWBR	8.1	8.1	7.3	6.1	5.2	4.4	4.0	4.1	4.9	6.0	7.5	8.1
	5% WB	MCDBR	10.4	10.0	9.7	9.5	9.6	9.3	8.7	7.8	6.7	6.8	9.0	10.2
		MCWBR	8.6	8.4	7.1	5.5	4.7	4.0	3.8	3.7	4.4	5.2	7.7	8.7
Clear-Sky Solar Irradiance	taub		0.351	0.386	0.437	0.499	0.511	0.426	0.434	0.433	0.429	0.433	0.370	0.347
	taud		2.342	2.211	2.042	1.889	1.890	2.158	2.166	2.173	2.170	2.135	2.314	2.371
	Ebn at Noon		884	878	851	809	798	864	855	855	848	816	855	870
	Edh at Noon		107	132	167	200	201	153	151	149	145	141	109	100
All-Sky Solar Radiation	RadAvg		4.12	5.08	6.41	7.33	7.95	8.43	8.19	7.67	6.75	5.51	4.43	3.85
	RadStd		0.17	0.20	0.17	0.18	0.21	0.08	0.10	0.10	0.14	0.14	0.11	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.36	N/A	N/A	+1.01	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	+0.45	N/A	N/A	+0.48	+0.46	+0.45	N/A	N/A	+217	+198

Nomenclature: See separate page